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1-1 PRIVATE TUTORS

Probability as Fractions, Decimals and Percentages

Name _____ Date _____ No calculator Version 1

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KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · event *(takahanga)* · equally likely *(tūpono ōrite)*

A Single events ●○○

Remember: Probability $= \frac{\text{number of outcomes you want}}{\text{total number of outcomes}}$, when every outcome is equally likely.

1. A fair dice is rolled. Write the probability of each event as a fraction in its simplest form.

a) Rolling a 4 _____

b) Rolling an even number _____

c) Rolling a number greater than 4 _____

d) Rolling a 7 _____

e) Rolling a number less than 7 _____

f) Rolling a prime number _____

B Fractions, decimals and percentages ●●○

Remember: A probability can be written as a fraction, a decimal or a percentage: $\frac{1}{4} = 0.25 = 25\%$. It is always between 0 and 1 (0% and 100%).

2. A bag holds 3 red, 5 blue and 2 green counters. One counter is taken out without looking. Write each probability as a fraction, a decimal and a percentage.

a) A red counter **b)** A blue counter **c)** A green counter

3. Cards numbered 1 to 20 are shuffled and one card is picked. Write each probability as a fraction, a decimal and a percentage.

a) The number 7 **b)** An even number **c)** A number greater than 15

C Events with more than one outcome ●●○

⚠ **Watch out:** Count every outcome that fits the event. For "a multiple of 3" on a spinner numbered 1 to 10, the outcomes 3, 6 and 9 all count.

4. A fair spinner is numbered 1 to 10. Find the probability of each event, as a fraction in its simplest form.

a) A multiple of 3 **b)** A prime number **c)** A number greater than 7

d) A factor of 10 **e)** An odd number less than 6 **f)** A square number

5. A bag holds 5 red, 4 blue, 2 green and 1 yellow counters. One counter is taken out without looking. Find the probability that it is:

a) red or blue **b)** green or yellow **c)** not red

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D Complementary events ●●○

> **Remember:** An event either happens or it does not, so the two probabilities add to 1 (or 100%). If the probability of rain is 0.3, the probability of no rain is $1 - 0.3 = 0.7$.

6. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.3 **b)** 35% **c)** $\frac{3}{8}$ **d)** 0.85

e) 8% **f)** $\frac{5}{12}$ **g)** 0.06 **h)** $\frac{1}{6}$

E Comparing likelihood ●●●

7. Choose the best word for each event: impossible, unlikely, even chance, likely or certain.

a) Getting a head when a fair coin is tossed

b) Rolling a 6 on a fair dice

c) Picking a red counter from a bag that holds only red counters

d) Rolling a number less than 6 on a fair dice

e) Rolling a 9 on a fair dice

8. Which is more likely? Show your working.

a) rolling a 6 on a fair dice, or picking a red counter from a bag with 2 red and 8 blue counters?

b) rolling an even number on a fair dice, or getting a head on a fair coin?

c) picking a vowel from the letters of MATHS, or picking a vowel from the letters of NUMBER?

★ Extension

9. A bag holds 10 counters. The probability of picking a red counter is $\frac{2}{5}$. How many red counters are in the bag?

10. The probability of winning a prize in a game is 0.2. About how many prizes would you expect to win in 40 games?

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Probability as Fractions, Decimals and Percentages

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KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · event *(takahanga)* · equally likely *(tūpono ōrite)*

A Single events ●○○

Remember: Probability $= \frac{\text{number of outcomes you want}}{\text{total number of outcomes}}$, when every outcome is equally likely.

1. A fair dice is rolled. Write the probability of each event as a fraction in its simplest form.

a) Rolling a 3 _____ **b)** Rolling an even number _____

c) Rolling a number greater than 2 _____ **d)** Rolling a 7 _____

e) Rolling a number less than 7 _____ **f)** Rolling a prime number _____

B Fractions, decimals and percentages ●●○

Remember: A probability can be written as a fraction, a decimal or a percentage: $\frac{1}{4} = 0.25 = 25\%$. It is always between 0 and 1 (0% and 100%).

2. A bag holds 4 red, 5 blue and 1 green counters. One counter is taken out without looking. Write each probability as a fraction, a decimal and a percentage.

a) A red counter **b)** A blue counter **c)** A green counter

3. Cards numbered 1 to 20 are shuffled and one card is picked. Write each probability as a fraction, a decimal and a percentage.

a) The number 7 **b)** An even number **c)** A number greater than 16

C Events with more than one outcome ●●○

⚠ **Watch out:** Count every outcome that fits the event. For "a multiple of 3" on a spinner numbered 1 to 10, the outcomes 3, 6 and 9 all count.

4. A fair spinner is numbered 1 to 10. Find the probability of each event, as a fraction in its simplest form.

a) A multiple of 4 **b)** A prime number **c)** A number greater than 6

d) A factor of 10 **e)** An odd number less than 6 **f)** A square number

5. A bag holds 4 red, 3 blue, 3 green and 2 yellow counters. One counter is taken out without looking. Find the probability that it is:

a) red or blue **b)** green or yellow **c)** not red

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D Complementary events ●●○

> **Remember:** An event either happens or it does not, so the two probabilities add to 1 (or 100%). If the probability of rain is 0.3, the probability of no rain is $1 - 0.3 = 0.7$.

6. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.4 **b)** 45% **c)** $\frac{5}{8}$ **d)** 0.65

e) 6% **f)** $\frac{7}{12}$ **g)** 0.08 **h)** $\frac{5}{6}$

E Comparing likelihood ●●●

7. Choose the best word for each event: impossible, unlikely, even chance, likely or certain.

a) Getting a tail when a fair coin is tossed

b) Rolling a 1 on a fair dice

c) Picking a blue counter from a bag that holds only blue counters

d) Rolling a number greater than 1 on a fair dice

e) Rolling a 0 on a fair dice

8. Which is more likely? Show your working.

a) rolling a 1 on a fair dice, or picking a red counter from a bag with 3 red and 9 blue counters?

b) rolling an odd number on a fair dice, or getting a tail on a fair coin?

c) picking a vowel from the letters of GRAPH, or picking a vowel from the letters of RANDOM?

★ Extension

9. A bag holds 12 counters. The probability of picking a red counter is $\frac{1}{4}$. How many red counters are in the bag?

10. The probability of winning a prize in a game is 0.25. About how many prizes would you expect to win in 60 games?

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